

Logistic Regression → ? imp + ? interview

ML → classification

1) Geometry 2) prob 3) logs - f^h ⇒ $f'()$

$D_n \rightarrow (LR)$

classifiers →

Blind classification

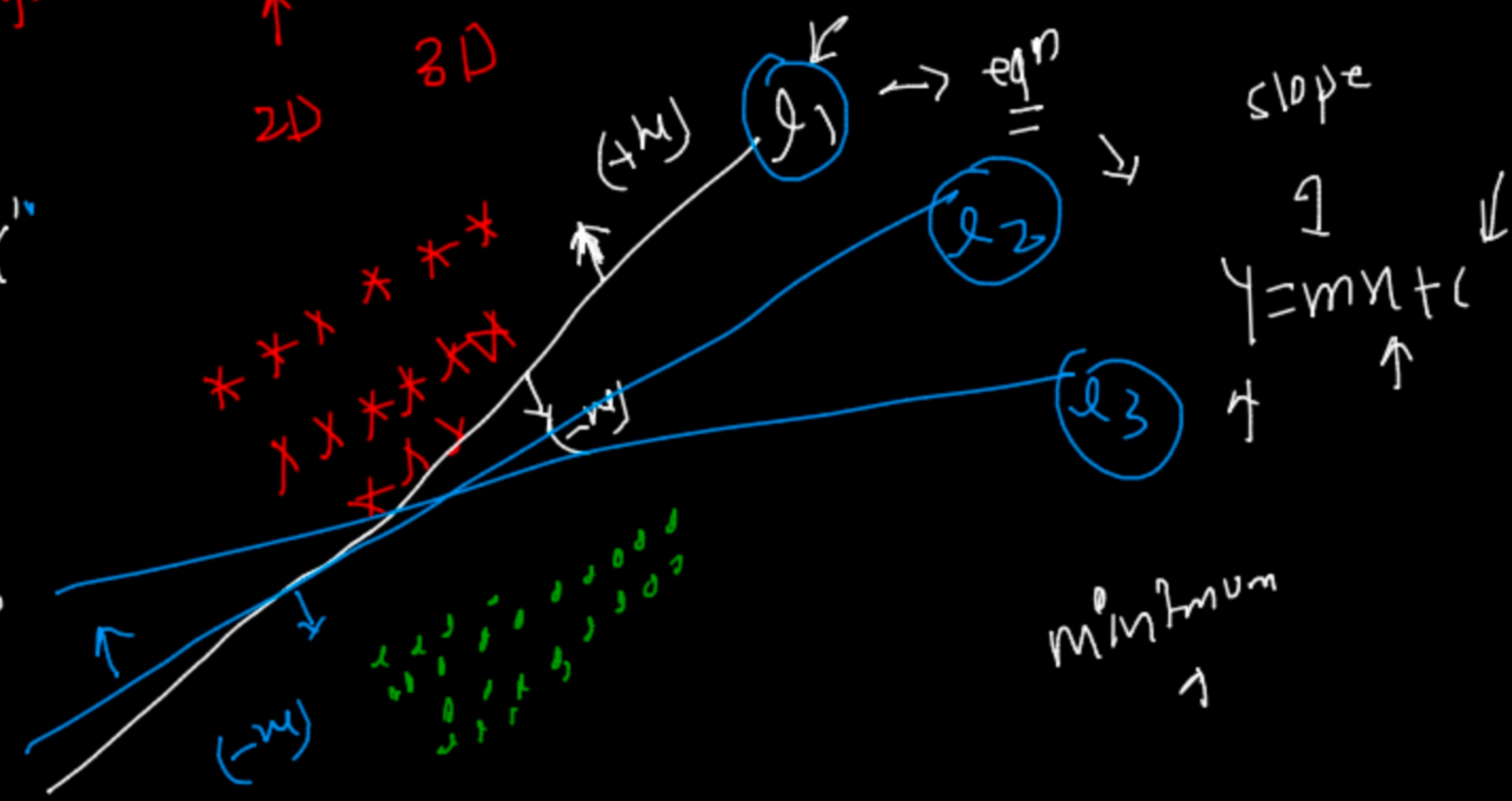
$(x, y) \rightarrow M()$
↳ $KNN()$

$\sqrt{KL}$ →

$f'(x) \rightarrow \hat{y}$

$P_n(x, y) \rightarrow \begin{cases} +ve \\ -ve \end{cases} \rightarrow LR() \rightarrow \text{line/plane/hyperplane}$

LR
separate → with
line



D_n

Big Assumption

↳ Classes are almost / perfectly linearly separable.

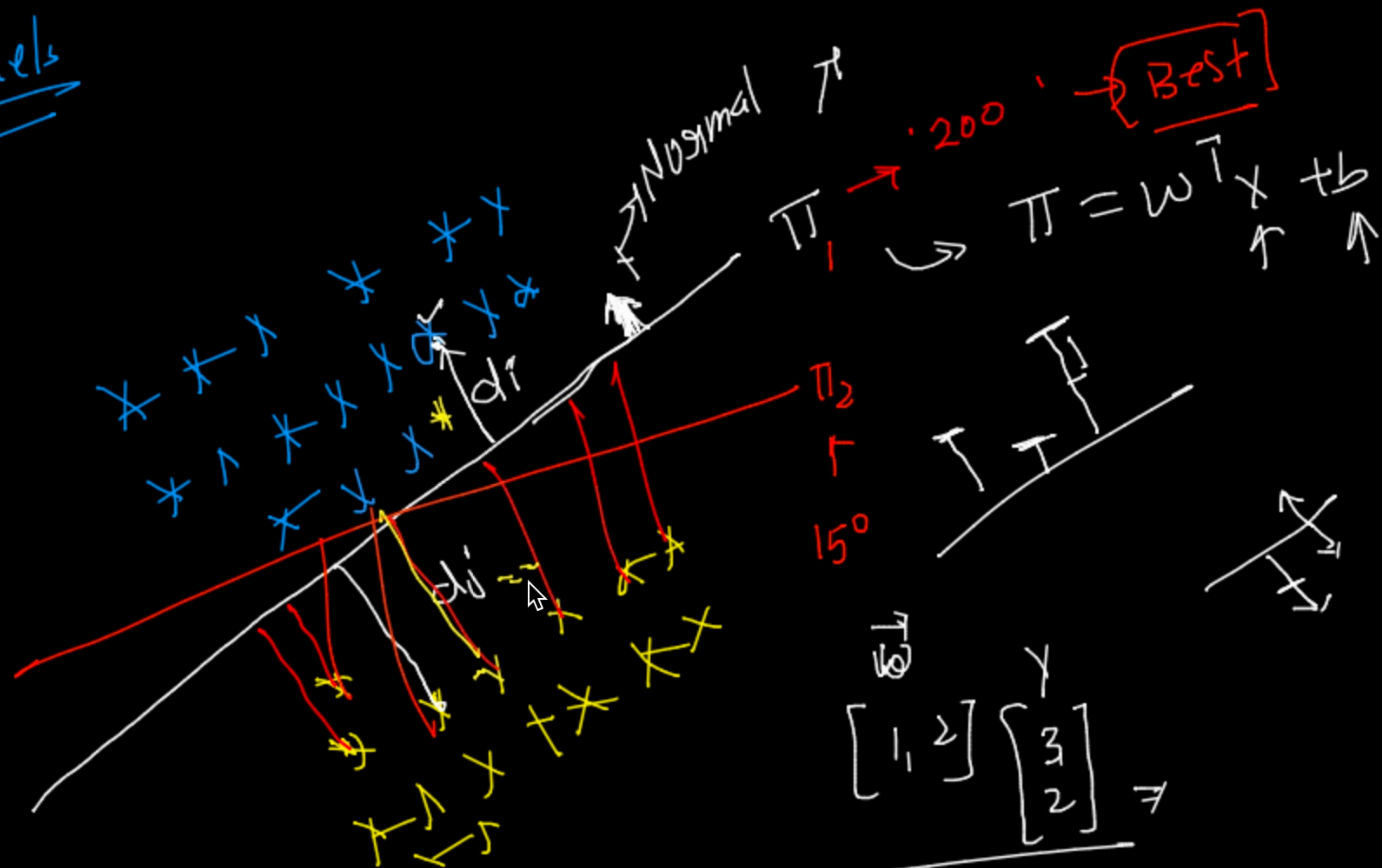
→ equation of plane $\pi = w^T x + b$ ← 'LA'

what we have $\Theta_n \{+w, -w\}$

→ task $\begin{bmatrix} w \\ b \end{bmatrix}$ find

our task is to find a plane $w^T x + b$

help



$$\frac{[1, 2] \begin{bmatrix} 3 \\ 2 \end{bmatrix}}{\sqrt{5}}$$

$w^T x_i$

$d_i = \frac{w^T x_i}{\|w\|} \geq 0$ [same side of normal]

$\|w\| \leftarrow \text{unit} = \|w\| = 1$

$$d_i = \frac{w^T x_i}{\|w\|} = \left| \frac{w^T x_i}{\|w\|} \right|$$

$D_n \rightarrow$ income, cs status

$x \rightarrow$ 14 650 yes $\rightarrow$

optimisation

hand

biased

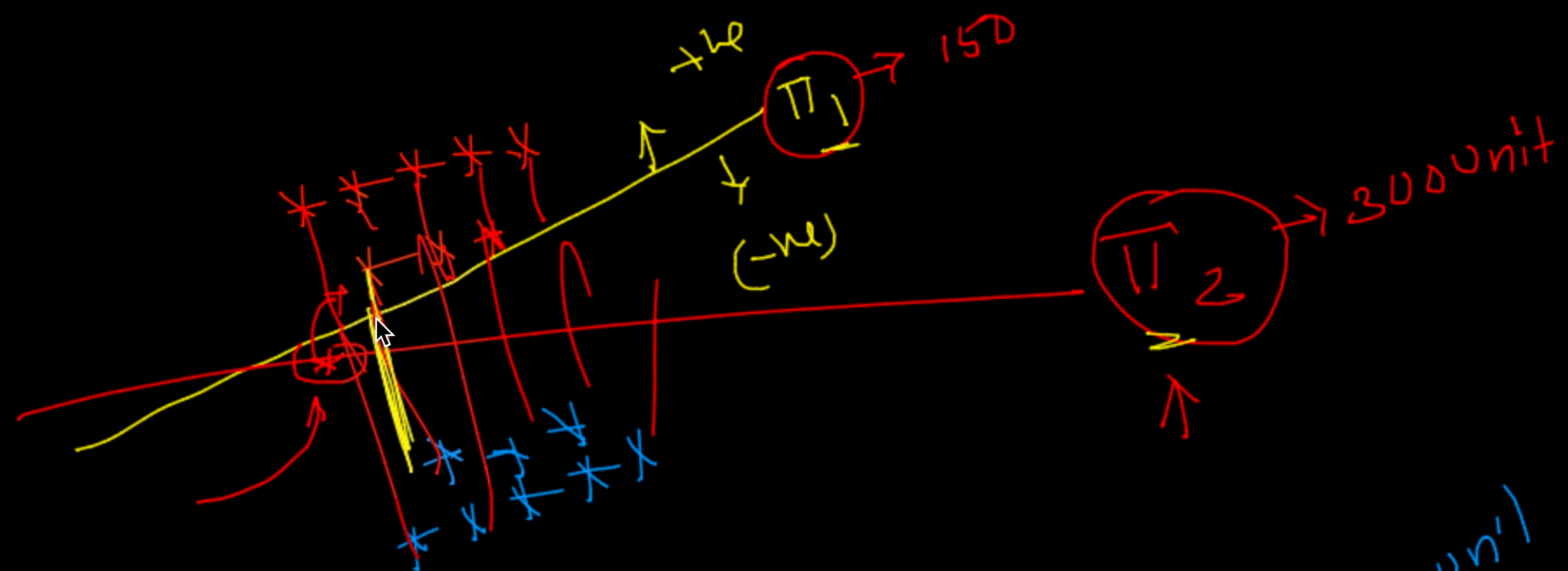
$$w_1 x_1 + w_2 x_2 + w_0 = 0$$

$\uparrow \quad \uparrow \quad \uparrow$

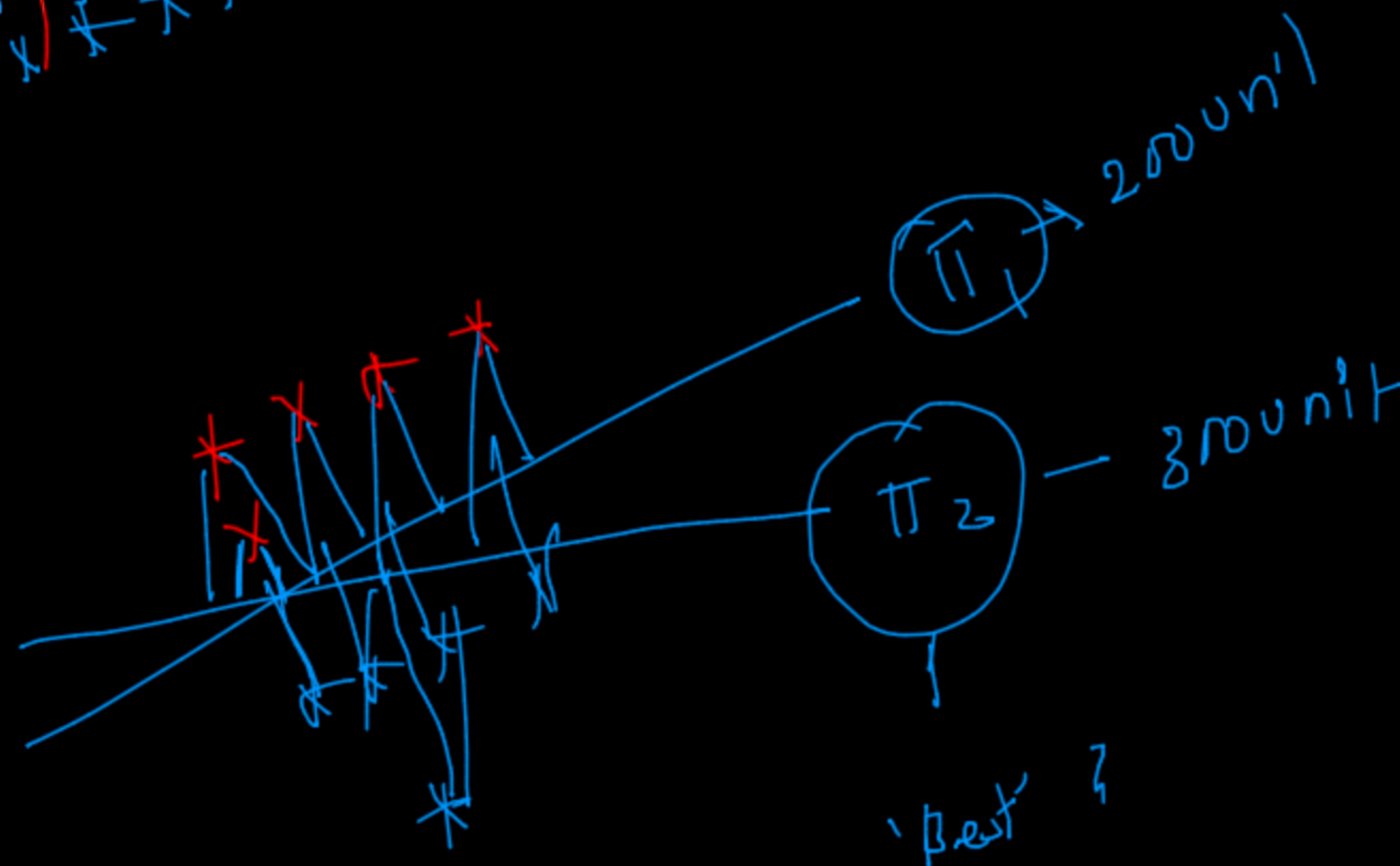
$14 * 0.4 + 660 * 0.1 + 2$

$\downarrow$

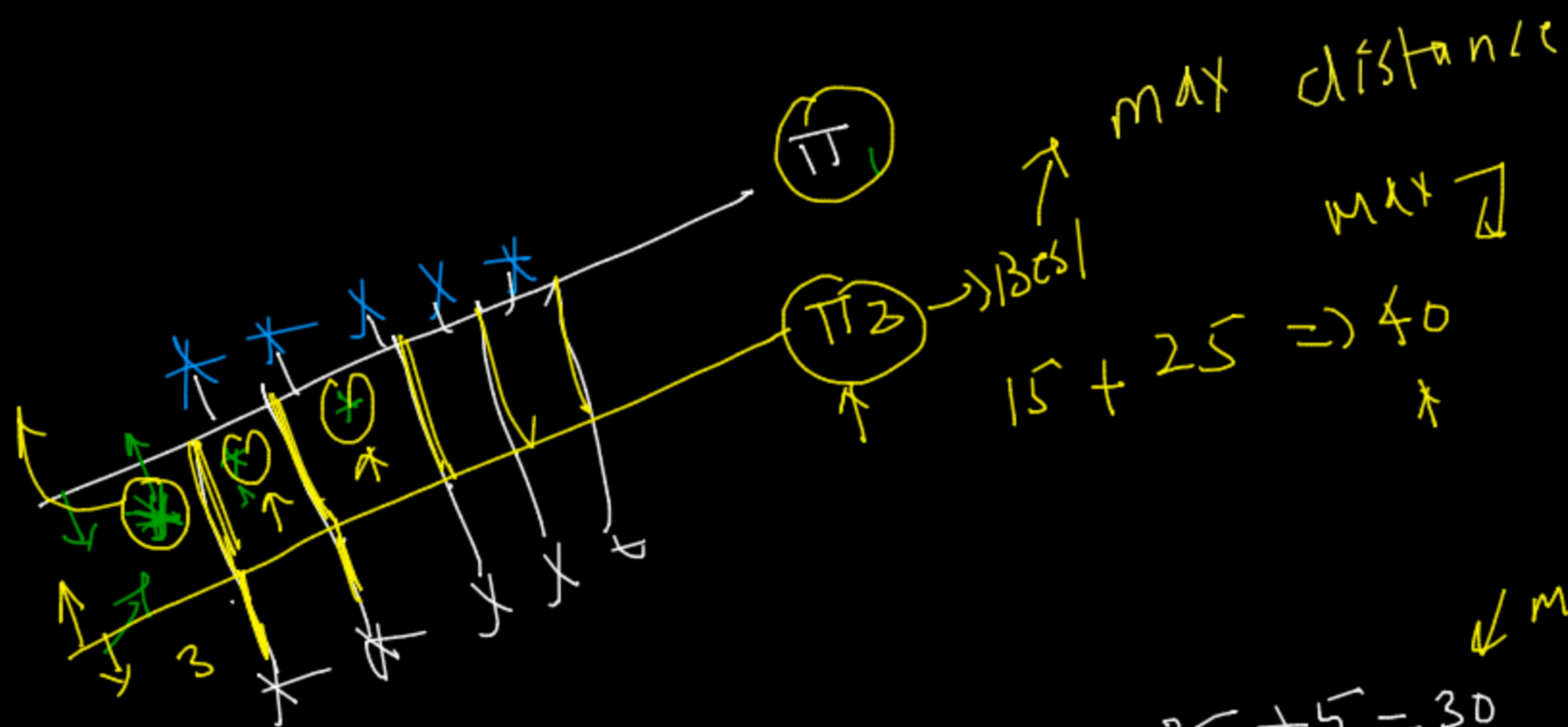
(ave)



$(-ve)$



correct



W/B

$y_i w^T x_i > 0 \rightarrow$ The LR model is correctly classified the pts

→

$\gamma_i w^T x_i > 0$ $\rightarrow$ The LR model is correctly classified the pts

$\rightarrow$

$y_i w^T x_i > 0 \rightarrow$ The LR model is correctly classified the pts

$\rightarrow$

$$\max \sum y_i w^T x_i$$

$\uparrow \quad \uparrow$
 $(x_i, y_i) \rightarrow \text{fixed}$

w it is only which will change

$\Rightarrow$ Task $\rightarrow \boxed{w} \leftarrow \text{Best to find} \rightarrow$

$$w^* = \underset{w}{\operatorname{argmax}} \left(\sum_{i=1}^n y_i w^T x_i \right)$$

$\uparrow$
 $f(x)$



Top limitations of problem

SUP Stochastic
Gradient
Descent

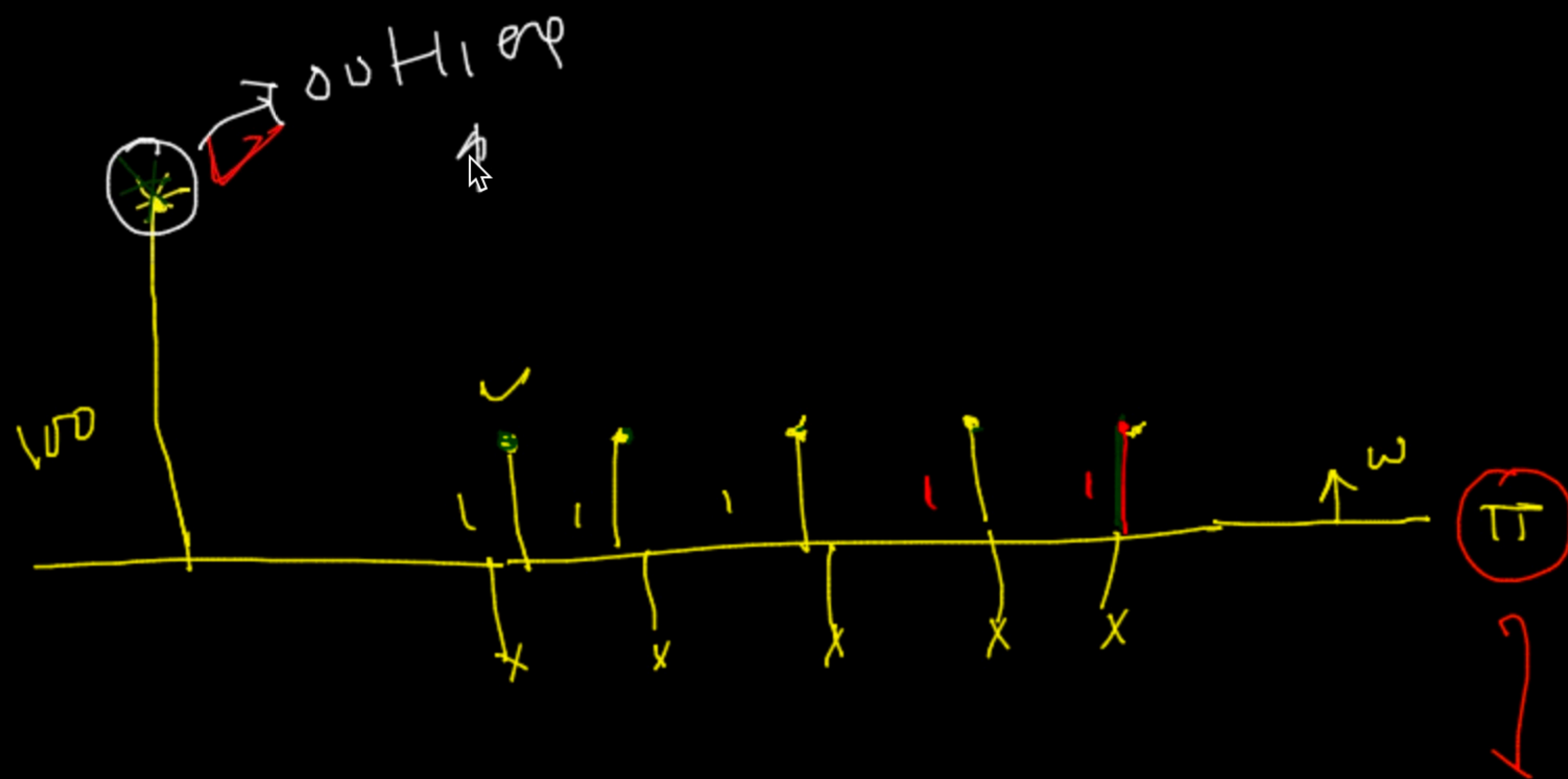
⇒ Squashing and sigmoid function

$$w^* = \underset{w}{\operatorname{argmax}} \sum_{i=1}^n y_i w^T x_i$$

$$\underset{y_i}{(1)} \underbrace{\underset{(+w)}{(1)}}_{w^T x_i} = 1$$

$$\frac{w^T x_i}{(-w)(+100)} \Rightarrow -100$$

↑
optimal
(w)
(use 1)
↑



10 good
11 ↑

4

$$\underline{7+1+1+1+1+5} - 100 = -90$$

↑

Sigmoid fn

↑

$$\arg \max \sum_{i=1}^n y_i w^T x_i$$

↓

$$\sigma(x) = \frac{1}{1 + e^{-x}}$$

$$\left[\begin{array}{l} \text{max-value} \rightarrow 1 \\ \text{min-value} \geq 0 \end{array} \right]$$

~~$\sigma(x)$~~

d ↑ large

$(w^T x_i)$

↑

squash

↓

0/1

$$w^* = \arg \max_w \sum_{i=1}^n \sigma(\gamma_i w^T x_i)$$

$$\text{i.e. } \arg \max_w \sum_{i=1}^n \frac{1}{1 + \exp(-\gamma_i w^T x_i)}$$

less impacted by outliers

11,

(0 -1)

1 1

$$w^* = \operatorname{argmax}_w \sum_{i=1}^n \log \left(\sigma(y_i w^T x_i) \right)$$

$$= \operatorname{argmax}_w \sum_{i=1}^n \log \left(\frac{1}{1 + \exp(-y_i w^T x_i)} \right)$$

$$= \operatorname{argmax}_w \sum_{i=1}^n \log 1 - \log(1 + \exp(-y_i w^T x_i))$$

$$= \operatorname{argmax}_w \sum_{i=1}^n -\log(1 + \exp(-y_i w^T x_i))$$

$$w^* = \operatorname{argmin}_w \sum_{i=1}^n \log(1 + \exp(-y_i w^T x_i))$$

↑
optimisation logistic regression

$$\log \frac{m}{n} \Rightarrow \log m - \log n$$

$$\frac{100}{2}$$

$$\log 100$$

$$100 \times \sqrt{\log 2}$$

Note

$$\operatorname{argmax}_x -f(x) \Leftrightarrow \operatorname{argmin}_x f(x)$$

